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Lecture notes for Applies statistics, Probability theory

Lecture 7.

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Presentation Overview

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- The **target population** is defined as the entire collection of objects or individuals about which we need some information. The target population must be well defined in terms of what constitutes membership in the population (e.g., income level, geographic area, etc.) and what characteristics of the population we are measuring (e.g., height, IQ, number of failures, etc.).
- A **random sample** of size n is said to be independent and identically distributed (iid) when the random variables $X_1, X_2, \dots, X_n$ each have a common distribution.
- There are two types of simple random sampling: **sampling with replacement** and **sampling without replacement**.
- Alternative sampling method: **stratified random sampling**, divides the population into levels, and then a simple random sample is taken from each level.



Definition

Let

$$X_1, X_2, \dots, X_n$$

be a random (iid) sample.

Let Ω be the background probability space. Then for $\omega \in \Omega$ the numbers

$$x_1 = X_1(\omega), x_2 = X_2(\omega), \dots, x_n = X_n(\omega)$$

are called sample realization.

Sample $\rightarrow$ theory

Sample realization $\rightarrow$ practice

Example: We weight 10 chocolate bars. The results are in grams:

$$x_1 = 100.1, x_2 = 99.7, \dots, x_{10} = 99.9$$

It's a sample realization.



Definition

Let $\omega \in \Omega$ be fixed.

Denote by

$$X_1^* \leq X_2^* \leq \dots \leq X_n^*$$

the increasing ordering of the sample realization $X_1(\omega), X_2(\omega), \dots, X_n(\omega)$. The random variables

$$X_1^* \leq X_2^* \leq \dots \leq X_n^*$$

are called order statistics (ordered sample).



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Empirical Cumulative Distribution Function (ECDF)

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Definition

Let $X_1^*, X_2^*, \dots, X_n^*$ the order statistics. Then the function

$$F_n^*(x) = \begin{cases} 0, & \text{if } x \leq X_1^*, \\ \frac{k}{n}, & \text{if } X_k^* < x \leq X_{k+1}^*, k = 1, \dots, n-1, \\ 1, & \text{if } x > X_n^* \end{cases}$$

is called the empirical distribution function. It is a step function jumping $\frac{1}{n}$ at any value of the sample. F_n^* is a random function.



The fundamental theorem of mathematical statistics.

$$\sup_{x \in \mathbb{R}} |F_n^*(x) - F(x)| \rightarrow 0, \text{ if } n \rightarrow \infty,$$

with probability 1.

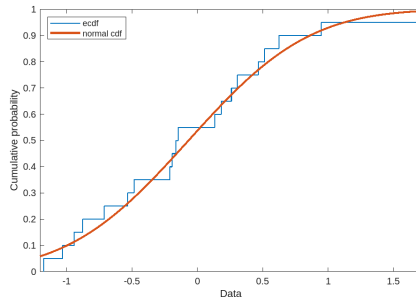


Fig. 1: ECDF and CDF fit



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Definition

A histogram is column diagram approximating the distribution of numerical data. Let $X_1, \dots, X_n$ be a sample. Divide the range of the sample into subintervals with the points $y_0 < y_1 < \dots < y_r$. The sample values should be in the interval (y_0, y_r) . The subintervals $[y_{i-1}, y_i)$ are called bins. Let k_i be the number of sample values falling into the bin $[y_{i-1}, y_i)$, $i = 1, \dots, r$. Then a rectangle is erected over the bin with area proportional to the frequency k_i , $i = 1, \dots, r$.



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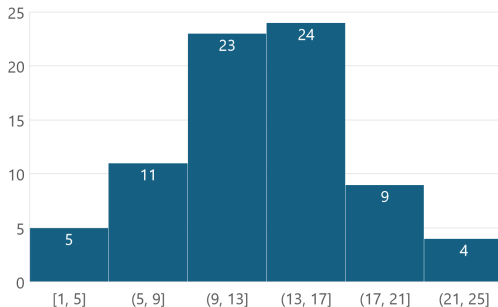


Fig. 2: Frequency histogram



Density histogram

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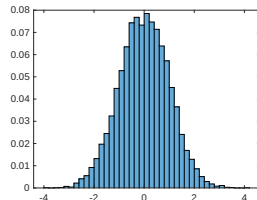
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If the cumulated area of the rectangles is n , then we obtain the frequency histogram. In this case the histogram is the real function f_n for which

$$f_n(x) = \begin{cases} \frac{k_i}{(y_i - y_{i-1})}, & \text{if } x \in [y_{i-1}, y_i), i = 1, \dots, r, \\ 0, & \text{otherwise.} \end{cases}$$

If the cumulated area of the rectangles is 1, then we obtain the density histogram. In this case the height of the i th rectangle is $\frac{k_i}{n(y_i - y_{i-1})}$. The density histogram is an approximation of the PDF.





Histogram plot

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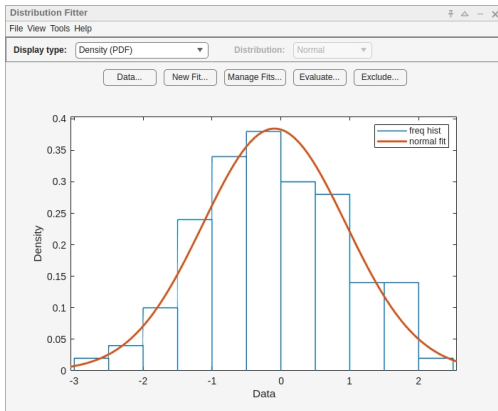


Fig. 3: MATLAB: **dfittool**



Kernel density estimation

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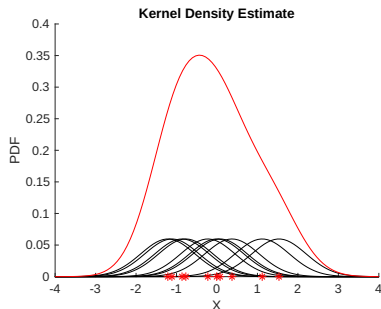
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Let $x_1, x_2, \dots, x_n$ be independent and identically distributed samples drawn from some univariate distribution with an unknown density f at any given point x . We are interested in estimating the shape of this function f . Its kernel density estimator is

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n K_h(x - x_i) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$$

where

K is the kernel — a non-negative function — and $h > 0$ is a smoothing parameter called the bandwidth or simply width.





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Empirical mean

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Let $X_1, X_2, \dots, X_n$ be sample for X . Then

$$\bar{X} = \frac{1}{n}(X_1 + X_2 + \dots + X_n)$$

is called the empirical mean. Assume that X has finite expectation $m = \mathbb{E}X$. In statistics the value of m is unknown. We can estimate it by $\bar{X}$. The expectation and the variance of $\bar{X}$ are

$$\mathbb{E}\bar{X} = \frac{1}{n} \sum_{i=1}^n \mathbb{E}X_i = m,$$

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_i^n \text{Var}(X_i) = \frac{\sigma^2}{n},$$

where $\sigma^2 = \text{Var}X$ is the theoretical variance.

Theorem: The empirical mean is an unbiased and consistent estimator of the theoretical expectation.



Empirical variance

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$$s_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

is called empirical variance. It is an estimator of the theoretical variance. A better estimator of the variance is the corrected empirical variance, that is

$$s_n^{*2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Steiner's formula will help us to calculate with $a = 0$:

$$ns_n^2 = (n-1)s_n^{*2} = \sum_{i=1}^n (X_i - a)^2 - n(\bar{X} - a)^2$$

Theorem: The empirical corrected empirical variance is an unbiased and consistent estimator of the theoretical variance.

Notation: The uncorrected empirical variance is biased, but asymptotically unbiased.



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Median, quantiles, quartiles

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Definition

μ is called the median of X if

$$P(X < \mu) < 1/2 \quad \text{and} \quad P(\mu > \mu) \leq 1/2$$

Definition

Let $0 < q < 1$. Then $Q(q)$ is called the q -quantile of X if

$$P(X < Q(q)) < q \quad \text{and} \quad P(X > Q(q)) \leq 1 - q.$$

By definition the q -quantile is $F^{-1}(q)$.

Remark: The 0.25-quantile (also as Q_1) is called the lower quartile.
The 0.75-quantile (Q_3) is called the upper quartile.



Estimating quantiles from a sample

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The empirical q -quantile $Q^*(q)$ approximates the theoretical quantile. Modern statistical packages rely on a number of techniques to estimate the quantiles. Here we show the technique used initially in Matlab.

$$h = Nq + 1/2 \quad Q^*(q) = X_{[h]}^* + (h - [h]) \cdot (X_{[h]+1}^* - X_{[h]}^*),$$

where $[h]$ is the integer part of h .

Example: Let's have the following sample realization in vector form: $x = [5, 3, 9, 7, 4]$. Then the ordered sample is the following :

$$x_1^* = 3, \quad x_2^* = 4, \quad x_3^* = 5, \quad x_4^* = 7, \quad x_5^* = 9.$$

Let's calculate the lower quartile (the 0.25-quantile):

$$h = 5 \cdot 0.25 + 0.5 = 1.75 \rightarrow [h] = 1$$

$$Q^*(0.25) = x_1^* + (1.75 - 1) \cdot (x_2^* - x_1^*) = 3 + 0.75 \cdot (4 - 3) = 3.75$$



Boxplot

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In descriptive statistics, a boxplot is a method for demonstrating graphically the locality, spread and skewness groups of numerical data through their quartiles. Outliers that differ significantly from the rest of the dataset may be plotted as individual points beyond the whiskers on the box-plot.

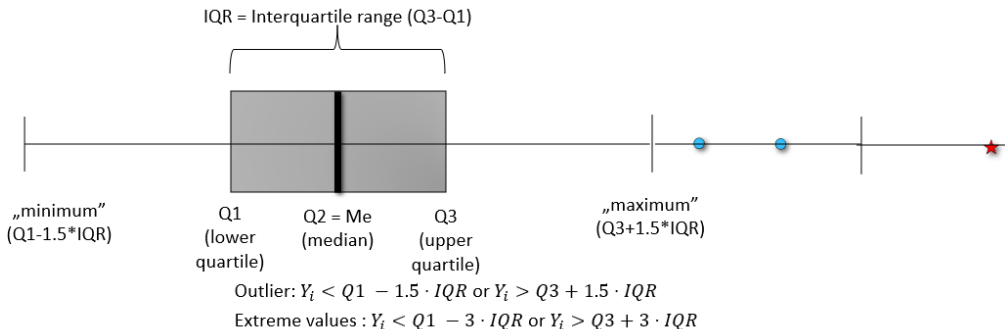


Fig. 4: Structure of a Boxplot



Q-Q plot

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A Q–Q plot (quantile–quantile plot) is a probability plot, a graphical method for comparing two probability distributions by plotting their quantiles against each other. A point (x, y) on the plot corresponds to one of the quantiles of the second distribution (y -coordinate) plotted against the same quantile of the first distribution (x -coordinate). This defines a parametric curve where the parameter is the index of the quantile interval.

If the two distributions being compared are similar, the points in the Q–Q plot will approximately lie on the identity line $y = x$. If the distributions are linearly related, the points in the Q–Q plot will approximately lie on a line, but not necessarily on the line $y = x$.



Q-Q plot

Remark: In Matlab the sorted elements in a vector are taken as the $0.5/n, 1.5/n, \dots, (n-0.5)/n$ quantiles.

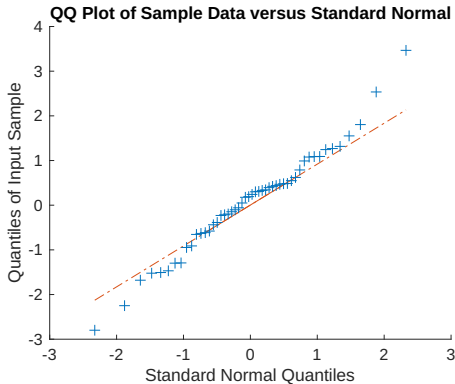


Fig. 5: QQ plot of a normal sample



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Maximum Likelihood Estimation

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A maximum likelihood estimator is the value of the parameter (or parameters) that maximizes the likelihood function of the sample. The likelihood function of a random sample of size n from density (mass) function $f(x, \theta)$ is the joint probability density (mass) function, denoted by

$$L(\theta; x_1, \dots, x_n) = f(x_1, \dots, x_n; \theta).$$

The equation provides the likelihood that the random variables take on a particular value $x_1, \dots, x_n$. Note that the likelihood function L is a function of the parameter θ , and that we allow θ to represent a vector of parameters. So the estimation of the parameter is

$$\hat{\theta} = \max_{\theta \in \Theta} L(\theta),$$

where Θ is the parameter space.



Maximum Likelihood Estimation

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Example. Let's assume we have an iid sample $x_1, \dots, x_n$ from the uniform distribution $\sim \text{Unif}(a, b)$. In this case the likelihood function of the parameters is

$$L(x_1, \dots, x_n; a, b) = f(x_1; a, b) \times \dots \times f(x_n; a, b) = \left(\frac{1}{b-a} \right)^n.$$

So for what parameters a and b can we maximize the L function?



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$$L(x_1, \dots, x_n; a, b) = f(x_1; a, b) \times \dots \times f(x_n; a, b) = \left(\frac{1}{b-a} \right)^n.$$

So for what parameters a and b can we maximize the L function?

To maximize L , we have to minimize $(b-a)$, where the smallest possible value of this difference according to the sample is $x_n^* - x_1^*$, because the $[a, b]$ interval has to cover the range of the sample. So

$$\hat{a} = x_1^*, \quad \hat{b} = x_n^*.$$

Notation. In most cases, to find the value $\hat{\theta}$ that maximizes the likelihood function, we take the derivative of L , set it equal to 0, and solve for θ .